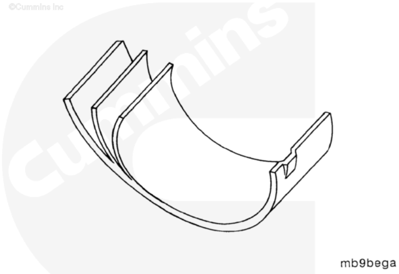
# Trainings

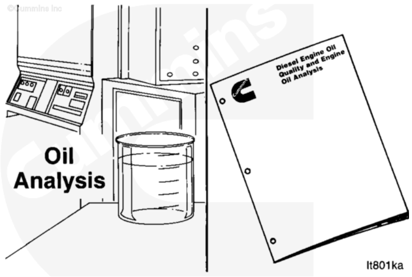
## General Information

The connecting rod bearings are a trimetal design with a steel backing.

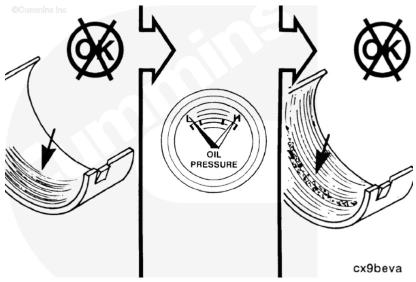



Improper maintenance of the lubrication system is the primary cause of reduced bearing life.

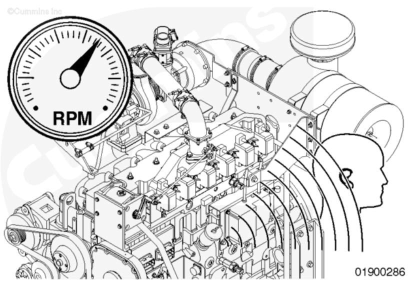
An oil analysis will aid in determining the extent of internal engine damage. For additional oil analysis, refer to Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340. (/qs3/pubsys2/xml/en/bulletin/3810340.html)



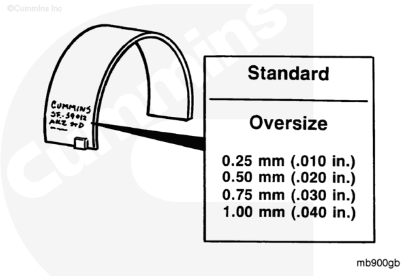

Normally, worn bearings can be detected by reduced oil pressure, but if this wear goes undetected, the excessive clearance will increase the impact between the bearing and crankshaft, causing a distinct knocking sound.




A connecting rod noise occurs when the engine is **not** loaded. Verify by first applying a load and then unloading and listening for the noise.



The connecting rod bearing shells are identified by steel-stamped characters on the back of the bearings. The characters indicate either standard (STD) or the amount oversize (OS).




## Preparatory Steps

**⚠ WARNING ⚠**

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

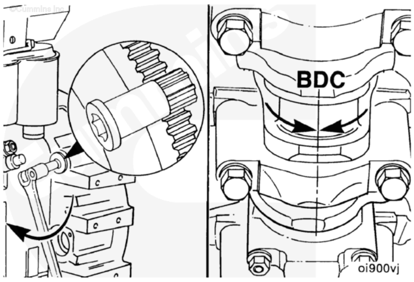
**⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.**

- Close the fuel supply valve. See equipment manufacturer service information.
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Remove the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-025.html)
- Remove the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-035.html)

## Remove

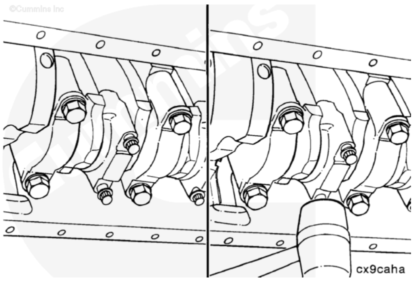
Rotate the crankshaft, using the engine barring gear, Part Number 3377371, to position two of the connecting rods at bottom dead center.




Loosen the connecting rod capscrew nuts.

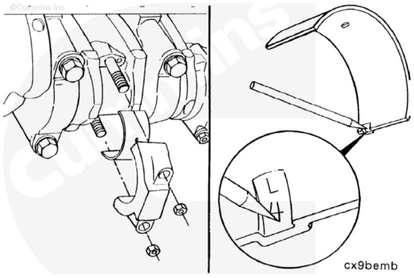
Do **not** remove the capscrew nuts.

Hit the connecting rod capscrew nuts with a plastic hammer to loosen the rod caps.




**⚠ CAUTION ⚠**

**The connecting rod bearings must be installed in the same location they were removed from if they are reused. Failure to install the connecting rod bearings in the same location will cause engine damage.**

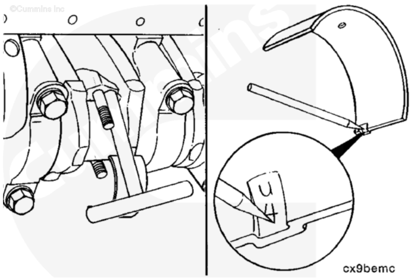



Remove the connecting rod capscrew nuts and rod cap.

Remove the bearing shell from the rod cap, and mark it as the lower bearing shell from the connecting rod number from where it was removed.

Push the connecting rod up far enough to allow the upper bearing shell to be removed.

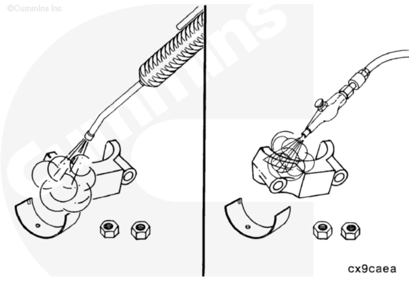
Remove the bearing shell, and mark it as the upper bearing shell and the connecting rod number from where it was removed.




## Clean and Inspect for Reuse

**⚠ WARNING ⚠**

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.




**⚠ WARNING ⚠**

**Some solvents are flammable and toxic. Read the manufacturer's instructions before using.**

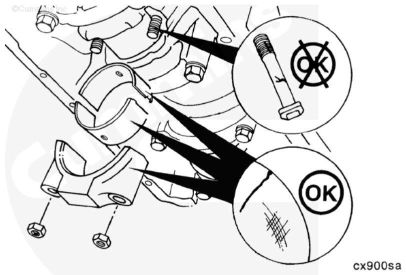
**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

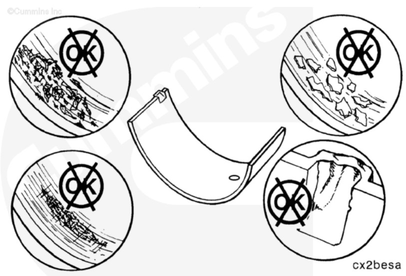
Use solvent and a soft bristle brush or a steam cleaner to clean the cap, capscrews, and bearings.

Dry the connecting rod cap, bearings, and capscrews with compressed air.

Inspect the connecting rod caps, connecting rod bearing saddles, and capscrews for nicks, cracks, burrs, scratches, or fretting.




Inspect the bearings for damage. Reference the Parts Reuse Guidelines, Bulletin 3810303 (/qs3/pubsys2/xml/en/bulletin/3810303.html), for information on bearing inspection.

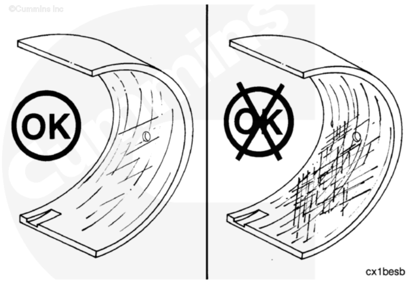



Replace any bearings with lock tang damage or scratches deep enough to be felt with a fingernail.

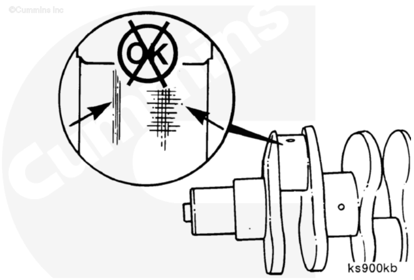
Replace any bearings that show pitting, flaking, or corrosion into the lining.

**Note : If bearings are damaged, they must be replaced as a set.**

**Note : Normal bearing wear produces a smooth finish that will wear into the lining. An exposed lining does not always indicate worn bearings. If large areas of the lining are visible in the bearings before the engine has accumulated 240,000 km [150,000 mi] or 3750 hours, inspect the engine for contamination from fine dirt particles, and correct the problem.**



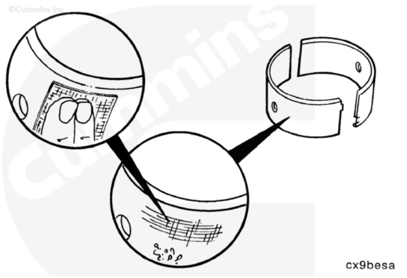

Inspect the crankshaft rod journals. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-006.html)




Inspect the bearing shell seating surface for nicks or burrs.

If nicks or burrs can **not** be removed with Scotch-Brite™ 7448, Part Number 3823258, the bearings **must** be replaced.

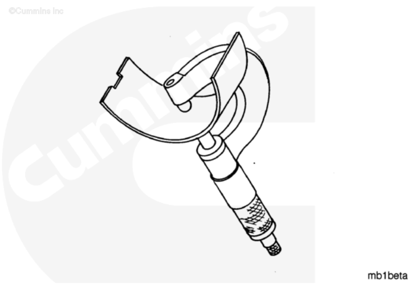
**Note : For more detailed information on bearing damage, reference Analysis and Prevention of Bearing Failures, Bulletin 3810387.**




Measure the rod bearing shell thickness with an outside micrometer that has a ball tip.

| Standard Connecting Rod Bearing Thickness (Used) | | |
|---|---|---|
| mm | | in |
| 2.43 | MIN | 0.096 |
| 2.47 | MAX | 0.097 |

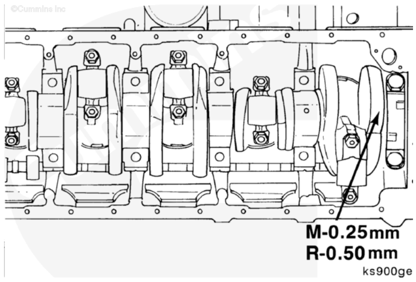
The bearing **must** be replaced if it is below the minimum specification.




**Note : Connecting rod bearings are identified with a part number and size stamped on the back.**

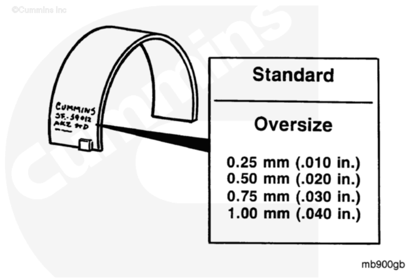
Determine the size of the removed rod bearing and obtain a set of new connecting rod bearings the same size.

**Note : Oversize service rod bearings are available for use with crankshafts that have been machined undersize. See the appropriate parts catalog.**



Crankshafts that are machined undersize in the connecting rod or main bearing journals are marked on the front counterweight. If the crankshaft is marked, check the bearing shell part number to make sure the correct bearing size is used.

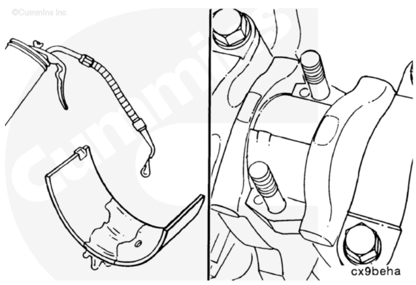
Bearing shells are identified by steel-stamped characters on the back of the bearings to indicate either standard (STD) or amount oversize (OS).




## Install

**⚠ CAUTION ⚠**

**The connecting rod bearings must be installed in the same location they were removed from if they are reused. Failure to install the connecting rod bearings in the same location will cause engine damage.**




Use clean engine oil to lubricate the crankshaft journal mating surface of the upper bearing shell.

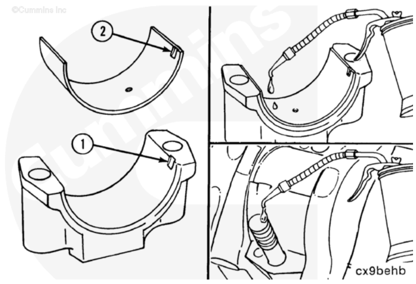
Install the upper bearing shell in the connecting rod with the tang of the bearing in the slot of the rod.

Pull the connecting rod against the crankshaft to hold the bearing in place.

Install the bearing shell in the connecting rod cap with the tang (2) of the bearing in the slot (1) of the cap.

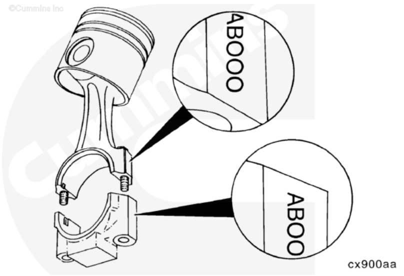
Use clean engine oil to lubricate the bearing shell to crankshaft journal mating surface.

Use clean engine oil to lubricate the threads of the connecting rod capscrews.



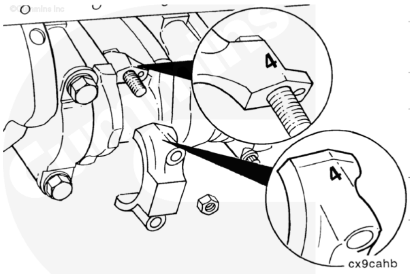

**Note : Unique serial numbers (not cylinder numbers) are stamped on the connecting rod and matching connecting rod cap.**

When the connecting rods and connecting rod caps are installed in the engine, the numbers on the connecting rods and connecting rod caps **must** match and be installed on the same side of the engine.




### ⚠ CAUTION ⚠

**The connecting rod cap number must match the number on the connecting rod and must be installed with the numbers aligned to prevent damage to the connecting rods and the crankshaft. The locking tang of the connecting rod cap must be toward the camshaft side of the cylinder block.**



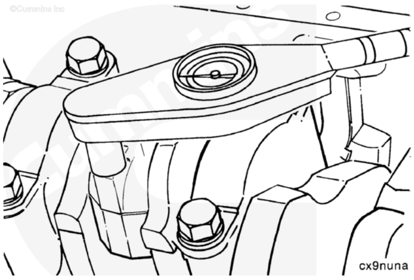

Lubricate under the connecting rod capscrew nuts with clean engine oil.

Install the connecting rod caps and rod capscrew nuts.

Tighten the connecting rod capscrew nuts in alternating sequence.

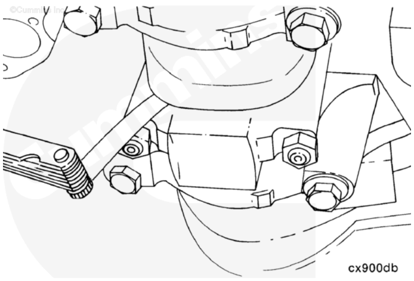
**Torque Value:**

1. 30 n•m [ 22 ft-lb ]
2. 70 n•m [ 52 ft-lb ]
3. 90° turn



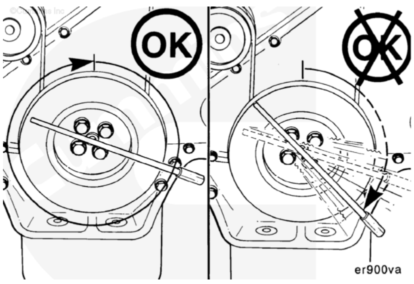

Measure the side clearance between the connecting rod and crankshaft.

| Connecting Rod Side Clearance | | |
|---|---|---|
| mm | | in |
| 0.10 | MIN | 0.004 |
| 0.33 | MAX | 0.013 |




**Note : The crankshaft must rotate freely.**

Check for freedom of rotation as the caps are installed. If the crankshaft does **not** rotate freely, check the installation of the rod bearings and the bearing size.



## Finishing Steps

⚠ **WARNING** ⚠

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Install the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-035.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-025.html)
- Fill the oil pan with clean engine oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Open the fuel supply valve. See equipment manufacturer service information.
- Operate the engine until the coolant temperature reaches 82°C [180°F]. Check for leaks and proper operation.

**Last Modified: 27-Mar-2019**